

Chapter 113 — Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Solifenacin 5 mg fails** → add mirabegron 50 mg (SYNERGY, BESIDE)
- **BESIDE:** combination > solifenacin 5 mg, **not** > solifenacin 10 mg
- **Mirabegron fails** → add antimuscarinic (MILAI II, 52 weeks)
- **Mirabegron add-on tested only with solifenacin**
- **Combination adds constipation;** no heart rate, blood pressure (BP), QT signal
- **Vibegron + antimuscarinic:** phase IIb only
- **Resistant nocturia** → α -blocker + desmopressin
- **DIGNITY:** onabotulinumtoxinA (onabotA) 200 U equal in anticholinergic users vs non-users
- **Systemic α -agonists for stress urinary incontinence (SUI):** US Food and Drug Administration (FDA) ban, December 2003
- **Intraprostatic onabotA = saline** in large randomised controlled trials (RCTs)

COMBINATION THERAPY

- **Rationale:** different mechanisms → use when antimuscarinic gives intolerable adverse events (AEs) or insufficient response
- **SYNERGY (12 weeks), SYNERGY II (12 months):** solifenacin 5 + mirabegron 50 > each monotherapy for incontinence (UI) and voids
- **BESIDE continence:** 46.0% combination vs 37.9% solifenacin 5 vs 40.2% solifenacin 10
- **BESIDE dry mouth:** highest with solifenacin 10 → combination = **alternative to dose escalation**
- **BESIDE age subanalysis:** benefit at ≥ 65 and ≥ 75 years
- **MILAI:** mirabegron 25 → 50 mg added after ≥ 4 weeks solifenacin 2.5/5 mg
- **MILAI II add-ons:** solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine extended release (ER) 4 mg
- **MILAI II titration:** dose doubled after 8 weeks, except tolterodine ER
- **Vibegron/tolterodine 100/4 mg** → ↓ micturitions, urgency vs monotherapies
- **Nocturnal polyuria (NP):** ~30% of men ≥ 60 years with benign prostatic hyperplasia (BPH)
- **α -blocker-resistant BPH + desmopressin** → $\geq 50\%$ fall in nocturnal voids in most
- **Desmopressin 0.1 mg oral, age ≥ 65 :** ↓ nocturnal voids and volume; no placebo control
- **Tolterodine after 12 weeks vaginal estradiol** → benefit; reverse order → none
- **Solifenacin $\pm$ vaginal oestrogen RCT:** no difference
- **Vaginal estradiol ring $\approx$ oral oxybutynin**
- **DIGNITY:** neurogenic detrusor overactivity (NDO), spinal cord injury (SCI) or multiple sclerosis (MS), 200 U
- **EMBARC:** idiopathic overactive bladder (OAB), 100 U, all anticholinergic-refractory → no user comparison

DRUGS THAT FAILED IN CLINICAL TRIALS TO IMPROVE STORAGE OR VOIDING FUNCTION

- **Vitamin D3:** ↓ urgency incontinence (UUI) only; obesity blunts effect
- **Elocalcitol 75/150 μ g (phase IIb):** ↓ incontinence; first-contraction volume unchanged
- **Calcium channel blockers (verapamil, nimodipine 30 mg):** no storage benefit
- **Cardiovascular calcium channel blockers:** no reported impaired bladder contractility
- **K⁺ channel openers:** underdosed to avoid cardiovascular AEs; ZD0947 25 mg null
- **Flupirtine (KCNQ/Kv7):** trial halted → **hepatotoxicity**

- **URO-902:** BK channel α -subunit gene therapy; phase 1 only
- **PSD503 (phenylephrine 20% vaginal gel):** ↓ pad weight; poor acceptance
- **Phenylpropanolamine:** non-selective → hypertension, haemorrhagic stroke
- **Clenbuterol 20 μ g twice daily:** tremor, tachycardia; unbalanced baselines
- **Botulinum toxin B:** ≤ 3 months at highest dose; ≤ 10 weeks at 5,000 U

DRUGS THAT FAILED TO IMPROVE VOIDING DYSFUNCTION

- **Prostatic onabotA RCT entry:** International Prostate Symptom Score (IPSS) >12, peak flow (Qmax) 5–15 mL/s; or IPSS >14, Qmax 4–15
- **Prostatic onabotA:** 100–300 U, no dose response; phase II urosepsis
- **PRX302:** prostate-specific antigen (PSA)-activated proaerolysin, 0.6 μ g/g
- **PRX302 entry:** IPSS ≥ 15 , Qmax ≤ 12 mL/s, prostate 30–100 mL
- **PRX302 at 3 months:** IPSS ↓ ~9, Qmax ↑ 3 mL/s; large placebo effect
- **NX-1207:** pro-apoptotic; phase III negative → suspended

CLASSIC TRAPS

- **Trap:** combination beats solifenacin 10 mg — **Actually:** BESIDE: only > solifenacin 5 mg
- **Trap:** any antimuscarinic + mirabegron is proven — **Actually:** only solifenacin tested
- **Trap:** antimuscarinic add-on to mirabegron untested — **Actually:** MILAI II, four agents
- **Trap:** combination raises BP or QT — **Actually:** no; constipation is the added AE
- **Trap:** vibegron combination established — **Actually:** phase IIb only
- **Trap:** estradiol added to tolterodine helps — **Actually:** only tolterodine added after estradiol
- **Trap:** onabotA weaker on anticholinergics — **Actually:** DIGNITY: equal
- **Trap:** EMBARK compares anticholinergic users — **Actually:** all refractory
- **Trap:** intraprostatic toxins treat BPH — **Actually:** = sham injection
- **Trap:** PSD503 failed on efficacy — **Actually:** failed on acceptability